

MENTORA ACADEMY

EXCELLENCE IN EDUCATION

Edexcel International A Level Chemistry (WCH14)

UNIT 4 — TOPICAL WORKSHEET PACK: 13A.2

EQUILIBRIUM CONSTANT, K_p

CANDIDATE NAME:	Usman
SUBJECT & CODE:	Chemistry Unit 4 — Rates, Equilibria & Organic (WCH14)
TOPIC AREA:	13: CHEMICAL EQUILIBRIA
SUB-TOPIC:	13A.2: Equilibrium Constant, K_p
TOTAL QUESTIONS:	4 Questions (Structured + MCQs + Calculations)
TOTAL MARKS:	6 Marks
EXAMINER FAQs:	3 Empirical Examiner FAQs & Traps Included

HOW TO USE THIS TOPICAL PACK:

1. Complete all structured questions under timed exam conditions.
2. Pay close attention to command words (Explain, Deduce, Calculate, State).
3. Review the complete worked mark scheme in Section C.
4. Study the Empirical Examiner FAQs in Section D to avoid common candidate traps.

SECTION A: TOPICAL PRACTICE QUESTIONS — 13A.2

Q1. 1. (a) Write the expression for the equilibrium constant, K_p , for the reaction: $\text{CO(g)} + 2\text{H}_2\text{(g)} \rightleftharpoons \text{CH}_3\text{OH(g)}$

[1 Marks]

WCH14/01/Jun23/Q4(a)

Q2. (b) In an experiment, a mixture of CO and H₂ was allowed to reach equilibrium at a total pressure of 50 atm. At equilibrium, the mole fractions were: CO = 0.15, H₂ = 0.35, CH₃OH = 0.50. Calculate the value of K_p and state its units.

[3 Marks]

WCH14/01/Jun23/Q4(b)

Q3. 2. What is the unit of K_p for the reaction $\text{H}_2\text{(g)} + \text{I}_2\text{(g)} \rightleftharpoons 2\text{HI(g)}$?

[1 Marks]

WCH14/01/Jan22/Q3

- A. atm
- B. atm⁻¹
- C. atm²
- D. No units

Q4. 3. Define the term partial pressure.

[1 Marks]

WCH14/01/Oct21/Q2

SECTION B: WORKED MARK SCHEME — 13A.2

Q	MARKING SCHEME & ACCEPTABLE ANSWERS	MARKS
Q1	Q1: 1. (a) Write the expression for the equilibrium constant, K_p , for the reaction: $\text{CO(g)} + 2\text{H}_2\text{(g)} \rightleftharpoons \text{CH}_3\text{OH(g)}$ $K_p = \frac{p_{\text{CH}_3\text{OH}}}{p_{\text{CO}} \times (p_{\text{H}_2})^2}$ Allow use of P or p for partial pressure. Do not allow square brackets [].	[1]
Q2	Q2: (b) In an experiment, a mixture of CO and H ₂ was allowed to reach equilibrium at a total pressure of 50 atm. At equilibrium, the mole fractions were: CO = 0.15, H ₂ = 0.35, CH ₃ OH = 0.50. Calculate the value of K_p and state its units. 1. Partial pressures: $p_{\text{CH}_3\text{OH}} = 0.50 \times 50 = 25 \text{ atm}$, $p_{\text{CO}} = 0.15 \times 50 = 7.5 \text{ atm}$, $p_{\text{H}_2} = 0.35 \times 50 = 17.5 \text{ atm}$ 2. $K_p = \frac{25}{(7.5 \times 17.5^2)}$ 3. = 0.0109 atm ⁻² (allow pa ⁻² if Pa used, but units must match)	[3]
Q3	Q3: 2. What is the unit of K_p for the reaction $\text{H}_2\text{(g)} + \text{I}_2\text{(g)} \rightleftharpoons 2\text{HI(g)}$? D	[1]
Q4	Q4: 3. Define the term partial pressure. The pressure a gas would exert if it alone occupied the whole container / total pressure $\times$ mole fraction.	[1]

SECTION C: EMPIRICAL EXAMINER FAQS & COMMON TRAPS — 13A.2**FAQ #1: Can I use square brackets [] for K_p expressions?****Category: Examiner Trap & Technique****COMMON EXAMINER TRAP / MISTAKE:**

Can I use square brackets [] for K_p expressions?

MODEL ANSWER / EXAMINER EXPECTATION:

No, using square brackets in a K_p expression will result in 0 marks. Square brackets mean concentration. You must use p() or P() to denote partial pressure.

FAQ #2: How do I calculate mole fraction?**Category: Examiner Trap & Technique****COMMON EXAMINER TRAP / MISTAKE:**

How do I calculate mole fraction?

MODEL ANSWER / EXAMINER EXPECTATION:

Mole fraction = (number of moles of the specific gas) / (total number of moles of all gases in the mixture). Make sure the sum of all mole fractions is exactly 1.

FAQ #3: What units should I use for total pressure?**Category: Examiner Trap & Technique****COMMON EXAMINER TRAP / MISTAKE:**

What units should I use for total pressure?

MODEL ANSWER / EXAMINER EXPECTATION:

You can use atm or Pa (or kPa), but you must be consistent. The unit of K_p will depend on the unit you used for pressure. Always check what unit the question provides.